



Entergy Operations, Inc.
River Bend Station
5485 U.S. Highway 61N
St. Francisville, LA 70775
Tel 225-381-4374

William F. Maguire
Site Vice President

RBG-47915

November 20, 2018

Attn: Document Control Desk
U.S. Nuclear Regulatory Commission
11555 Rockville Pike
Rockville, MD 20852-2738

Subject: Licensee Event Report 50-458 / 2018-008-00
River Bend Station, Unit 1
Docket No. 50-458
License No. NPF-47

Dear Sir or Madam:

In accordance with 10 CFR 50.73, enclosed is the subject Licensee Event Report. This document contains no commitments. If you have any questions, please contact Mr. Tim Schenk at 225-381-4177.

Sincerely,

A handwritten signature in cursive script, appearing to read "WFM", written in dark ink.

WFM/twf

Enclosure

cc: (with Enclosure)

U.S. Nuclear Regulatory Commission
Attn: Ms. Lisa M. Regner, Project Manager
09-D-14
One White Flint North
11555 Rockville Pike
Rockville, MD 20852

U.S. Nuclear Regulatory Commission
Region IV
1600 East Lamar Blvd.
Arlington, TX 76011-4511

RBG-47915

Page 2 of 2

NRC Senior Resident Inspector
Attn: Mr. Brian Parks
5485 U.S. Highway 61, Suite 1
St. Francisville, LA 70775

Public Utility Commission of Texas
Attn: PUC Filing Clerk
1701 N. Congress Avenue
P. O. Box 13326
Austin, TX 78711-3326

INPO
(via ICES reporting)

RBf1-18-0213

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name River Bend Station - Unit 1	2. Docket Number 05000 458	3. Page 1 OF 3
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4. Title Potential Loss of Safety Function and Unanalyzed Condition due to Legacy Design Deficiency in Control Building HVAC.

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
09	25	2018	2018	08	00	11	20	2018	NA	05000 NA

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☒ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
90	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER
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Licensee Contact Tim Schenk, Manager - Regulatory Assurance	Telephone Number (Include Area Code) 225-381-4177
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13. Complete One Line for each Component Failure Described in this Report
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Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

14. Supplemental Report Expected	15. Expected Submission Date	Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No		NA	NA	NA

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)
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On September 25, 2018, with the plant operating at 90 percent power, a design vulnerability in the Control Building Ventilation (HVK) system control logic was identified during an extent of condition review of a previously reported condition. The review identified a vulnerability in the HVK Chiller start logic. The design vulnerability had been in place since original plant construction and could cause a loss of all Control Building Cooling for up to 20 minutes following a Loss of Offsite Power. Compensatory measures were put in place to restore operability of the system until a temporary modification could be installed to remove the design vulnerability. This condition was of minimal significance with respect to the health and safety of the public.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
River Bend Station - Unit 1	05000- 458	YEAR 2018	SEQUENTIAL NUMBER 08	REV NO. 00

NARRATIVE**BACKGROUND**

The Control Building Ventilation (HVK) [VI] system consists of two 100-percent capacity Chillers (**CHU**) which provide cooling to each ventilation system sub-loop. Each of which provides cooling, heating, ventilation, pressurization, and smoke removal for several areas within the building. The ventilation system comprises two independent, redundant trains of components and subsystems either of which supports the safety function of providing a controlled environment in the main control room and the environmental requirements of the safety-related electrical equipment in the building. Each HVK subsystem is supported by the respective safety-related emergency diesel generator and conforms to the single failure criterion.

The HVK System provides chilled water to the cooling coils in three separate air handlers within each division. The air handling system (HVC) removes the heat generated within the main control room, electrical equipment rooms and chiller rooms. Each air handler serves both trains of their assigned areas (e.g., both the Division I and II standby switchgear rooms cooling may be fully served by the air handler in either train).

Normally, one HVK Chiller is in operation, with one from the opposite division in standby. This provides for an immediate and automatic swap with no manual actions should a problem exist in the operating division. Division I HVK consist of HVK Chillers 'A' and 'C' and Division II HVK consist of HVK Chillers 'B' and 'D'.

REPORTED CONDITION

On September 25, 2018, with the plant operating at 90 percent power, Engineering identified that if either HVK Chiller 'A' or HVK Chiller 'B' was started up to 20 minutes prior to a Loss of Offsite Power (LOP) event then any single failure in the opposite division that causes that division not to run would cause a loss of all Control Building cooling until the HVK Chiller 20 minute quick restart timer times out. Additionally, if either the Division I or the Division II HVK start sequence is initiated (manually or automatically) before the 20 minute time out, then the associated chilled water pump would start and then trip as the Chiller would not start in the required time period. In this condition, HVK Chiller 'A' and HVK Chiller 'B' cannot be used as the running or standby Chiller. When this condition exists, if either of these Chillers is started then the Control Building cooling system does not meet single failure criteria until the 20 minute quick restart timer times out.

HVK Chiller 'C' and HVK Chiller 'D' have not been affected by this deficiency since 2016 when a digital control system upgrade was installed. The new control system changed the allowed number of starts in a 20 minute period from 1 to 2.

This event is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B), 10 CFR 50.73(a)(2)(ii)(B) and 10 CFR 50.73(a)(2)(v)(D).

CAUSAL ANALYSIS

The HVK control logic design deficiency was introduced during original plant design and construction. The deficiency was previously identified several times over the life of the plant. However, the deficiency was not recognized as a violation of licensing basis requirements and corrective actions were not put in place to remediate the deficiency.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
River Bend Station - Unit 1	05000-458	YEAR 2018	SEQUENTIAL NUMBER 08	REV NO. 00

NARRATIVE

CORRECTIVE ACTION TO PREVENT RECURRENCE

Compensatory measures were put in place so that only HVK Chiller 'C' or HVK Chiller 'D' could be the running or standby Chiller until a temporary modification was put in place to bypass the Quick Restart timer in the HVK Chiller 'A' and HVK Chiller 'B' start logic. If either HVK Chiller 'A' or HVK Chiller 'B' was already running and had been running for longer than 20 minutes, then it was allowed to continue to run.

A temporary modification to bypass the quick restart timer in the HVK Chiller 'A' and HVK Chiller 'B' start logic was installed in October 2018. Administrative controls were put in place to restrict operators from unnecessarily restarting the Chillers within 20 minutes of a previous start.

Procedure revisions were made to restrict operators from restarting a Chiller before 20 minutes have elapsed from the previous start. An additional procedure change was made so that following a LOP, if a Chiller is restarted within 20 minutes of the previous start, that Chiller must be run for 2 hours or secured for 4 hours to allow the motor to cool before another start attempt.

An extent of condition for HVK single failure vulnerabilities and design deficiencies was performed. No new vulnerabilities were identified.

A permanent modification will be performed to eliminate the HVK control logic deficiency and is being tracked by the Corrective Action Program.

PREVIOUS OCCURRENCE EVALUATION

A similar HVK control logic vulnerability event was identified in April 2018 and reported by LER-2018-001 on June 7, 2018.

SAFETY SIGNIFICANCE

In the unlikely event of a LOP within 20 minutes of an HVK Chiller 'A' or HVK Chiller 'B' start and a single failure in the opposite division, control room operators would have responded to the event with established procedures for a loss of Control Building Ventilation to restore the safety function with either HVK Chiller 'C' or HVK Chiller 'D'. Thus, this event was of minimal significance to the health and safety of the public.

(NOTE: Energy Industry Identification System component function identifier and system name of each component or system referred to in the LER are annotated as (**XX**) and [XX], respectively.)